

Cellula Technologies - RNN Training Results

One Week Task - Multiclass Text Classification

93.33%	0.9344	0.9330	≥ 0.75 (PASS)
Accuracy	Macro F1	Weighted F1	Target

Executive Summary

The submitted RNN uses a Bidirectional SimpleRNN over trainable word embeddings. The model was evaluated on a 300-row stratified test split. It achieved a Macro F1 of **0.9344**, which exceeds the assignment target of **0.75**.

Training used EarlyStopping and ReduceLROnPlateau. The best validation-loss weights were restored from epoch 4; training stopped at epoch 9.

Dataset and Preprocessing

- Dataset size: 3,000 rows and 9 target classes.
- Input features: query text + image description text.
- Split: 80% train (2,400), 10% validation (300), 10% test (300), stratified by class.
- Vocabulary size: 3,972 tokens; padded sequence length: 44.
- Class imbalance was handled with clipped class weights in the range 0.6 to 3.0.
- The tokenizer was fitted on the training split only.

Data Quality Findings

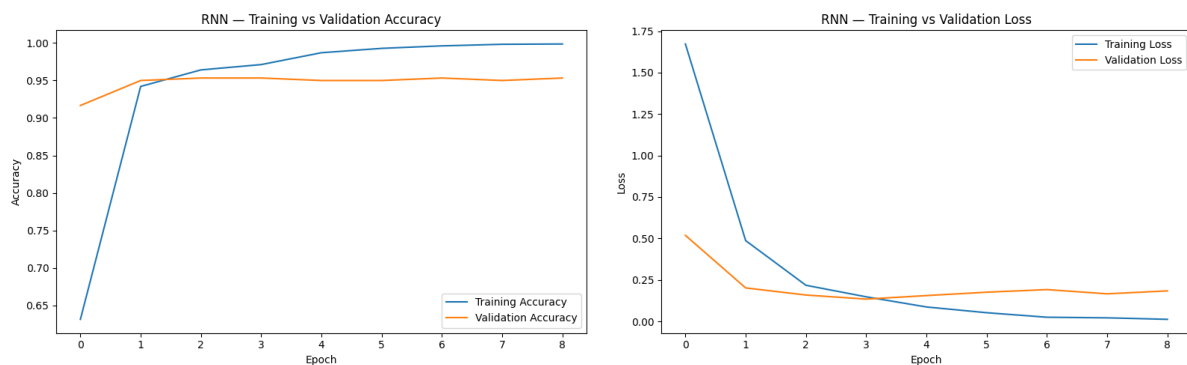
Finding	Observed value
Exact duplicate rows	973
Rows in repeated-query groups	1,000
Known template-like rows	1,000 (33.33%)
Unique image descriptions	12
Validation exact overlap with Train	90 / 300 (30.00%)
Test exact overlap with Train	114 / 300 (38.00%)

Interpretation: The benchmark result can be optimistic because repeated/template-like inputs can occur in both training and test partitions. The reported score is therefore best interpreted as performance on the supplied assignment benchmark rather than a leakage-free estimate on completely novel production data.

RNN Architecture

Layer	Output Shape	Parameters
Embedding	(None, 44, 128)	508,416
SpatialDropout1D	(None, 44, 128)	0
Bidirectional SimpleRNN	(None, 44, 256)	65,792
GlobalMaxPooling1D	(None, 256)	0
Dense (ReLU)	(None, 128)	32,896
Dropout	(None, 128)	0
Dense (Softmax)	(None, 9)	1,161
Total		608,265

Training Curves



The model converged quickly. Validation performance stabilized earlier than training performance, and EarlyStopping restored the best validation-loss checkpoint.

Training Configuration

Setting	Value
Optimizer	Adam
Initial learning rate	0.0005
Batch size	32
Maximum epochs	35
EarlyStopping patience	5
ReduceLROnPlateau factor	0.5
Loss	Sparse categorical cross-entropy

Final Test Results

93.33%	0.9344	0.9330	>= 0.75 (PASS)
Accuracy	Macro F1	Weighted F1	Target

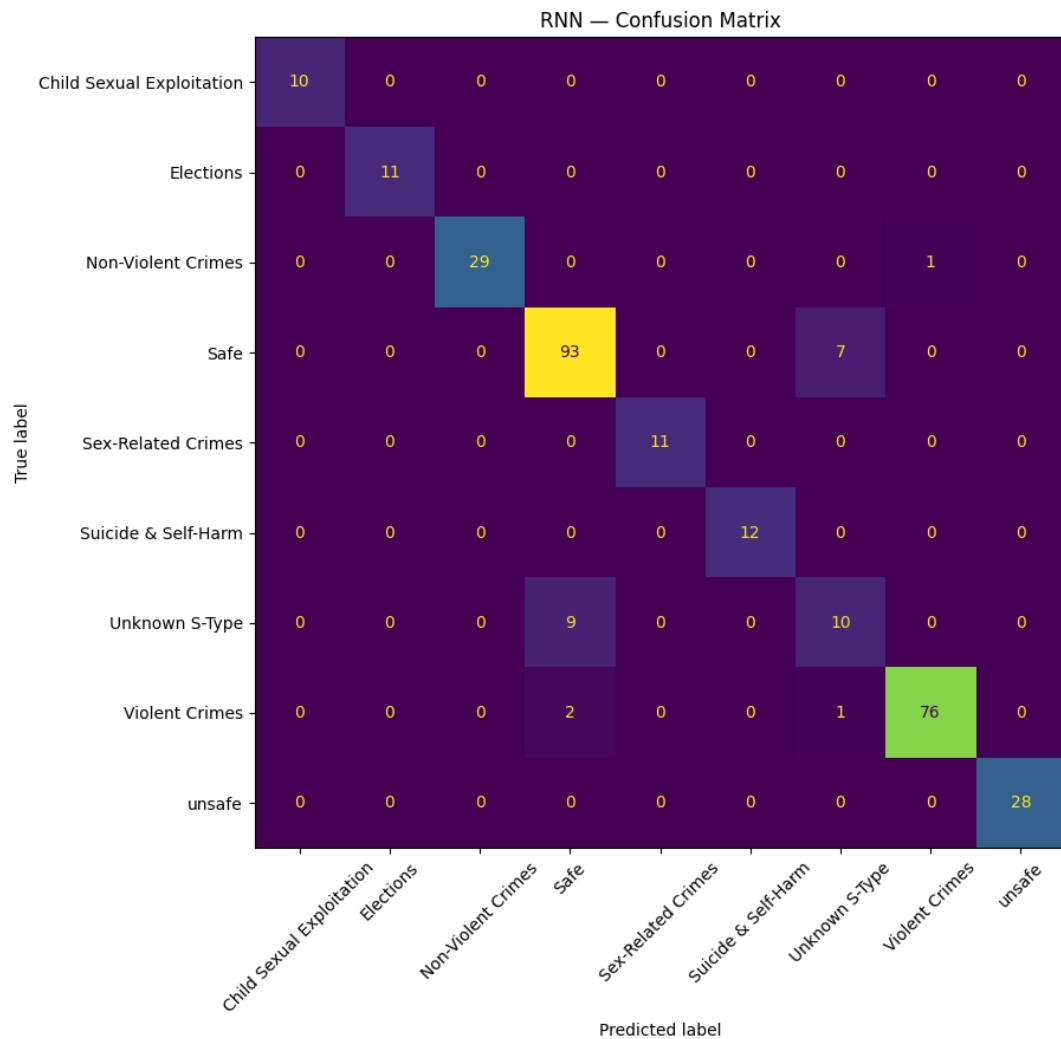
Per-Class Classification Report

Class	Precision	Recall	F1	Support
Child Sexual Exploitation	1.0000	1.0000	1.0000	10
Elections	1.0000	1.0000	1.0000	11
Non-Violent Crimes	1.0000	0.9667	0.9831	30
Safe	0.8942	0.9300	0.9118	100
Sex-Related Crimes	1.0000	1.0000	1.0000	11
Suicide & Self-Harm	1.0000	1.0000	1.0000	12
Unknown S-Type	0.5556	0.5263	0.5405	19
Violent Crimes	0.9870	0.9620	0.9744	79
unsafe	1.0000	1.0000	1.0000	28

Result Interpretation

The lowest per-class F1 is for **Unknown S-Type** at **0.5405**. This class remains the main source of classification difficulty. The RNN nevertheless exceeds the required F1 threshold by a wide margin.

Confusion Matrix



Conclusion

The RNN model **passes** the assignment requirement with Macro F1 = **0.9344** versus the required **0.75**. For production use, a manually verified and deduplicated dataset, group-aware splitting, and a genuinely unseen external holdout set are recommended.